

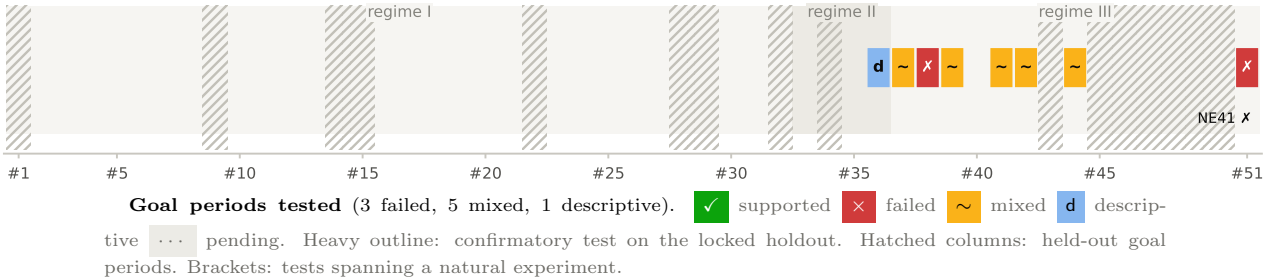
H109: The forced erasure is a demagnetizing pulse: where is the room's order stored?

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1.5/5 Usefulness 2/5

Bottom line. A forced context erasure does not demagnetize an agent's room alignment ($\delta_F = -0.08 [-0.23, 0.07]$, power 1.00 at a 30% drop); room order lives in the agent's work, prompt and memory, not in its context window.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



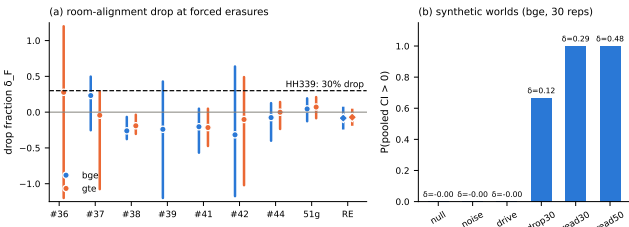
WHAT HAPPENED

HH339 says that a forced context erasure (NE41, the 41-call cap) is a demagnetizing pulse. If a room's content order is held by coupling through the context window, an agent's alignment with its room's direction should drop right after an erasure and recover as it re-reads the room. H109 projects every statement on the leave-agent-out spontaneous room axis (H100's Δ_{spont}) in 8 non-holdout regime-III two-room periods and compares 3,477 forced erasures with gap-matched within-segment boundaries of the same agent and day. **There is no pulse.** Pooled $\delta_F = -0.08 [-0.23, 0.07]$ (gte -0.07), where a 30% drop is detected with power 1.00. HH339's kill fires. The holdout is not run.

PHYSICAL MODEL

Model 11 (soft-spin $O(32)$ in a two-domain room system) with the erasure as a natural scramble of one channel (model 04). A statement is $\tilde{\mathbf{x}}_t = \mathbf{a}_i + \sigma_r m(t) \mathbf{u}_P + \boldsymbol{\eta}_t$. Context-held order: m drops by δ after an erasure and recovers with the room items R read. Self-made field (own work, prompt, memory): $\delta = 0$. The kickoff, held in the prompt since NE13, must not move in either case.

$$a_t = \sigma_{r(t)} \langle \tilde{\mathbf{x}}_t - \hat{\mathbf{a}}_i, \hat{\mathbf{u}}_i \rangle, \quad m(R) = m_\infty [1 - \delta e^{-R/R_0}]$$
$$\delta_F = - \frac{\sum_s n_{F,s} (\bar{D}_{F,s} - \bar{D}_{W,s})}{\sum_s n_{F,s} \bar{A}_{\text{pre},F}}, \quad D = \bar{a}_{\text{post}} - \bar{a}_{\text{pre}}$$



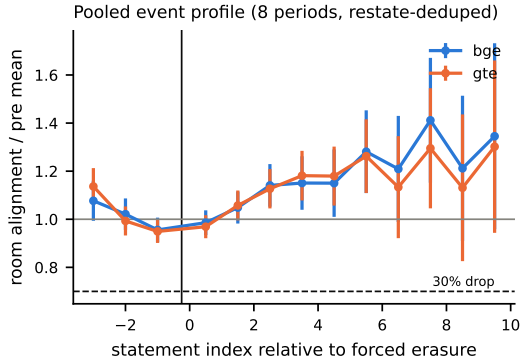
Observables. (a) Drop fraction δ_F per period (bge blue, gte orange; 95% agent-day bootstrap CIs, clipped at ± 1.2) and the random-effects mean (diamonds). The dashed line is HH339's 30% drop. Periods whose pre-erasure alignment is not identified are omitted. (b) Real-skeleton synthetic worlds: probability that the pooled CI excludes 0, with each world's noise-free estimand. The null, erasure-noise and common-drive worlds never reject; the 30% world always does.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator: resetting an agent's context does not erase its room's content identity. Alignment with the room stays within -7% to $+23\%$ of its pre-erasure level (pooled 95% CI). Use erasures to break loops or clear clutter (H44, H69) without worrying about room differentiation. To change what a room talks about, change its work or instructions. Reading the room has only a small pull ($\kappa_R \approx 0.02$ per e-fold of items read).

RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
P0 size ≤ 0.07 ; power ≥ 0.8	size 0.00; power 1.00 pooled, 2/8 mostly periods	mostly
P1 $\delta_F \geq 0.30$ (HH339); kill	$-0.08 [-0.23, 0.07]$; gte $-0.07 [-0.18, 0.04]$	killed
P2 $ \delta_K < 0.15$	$0.08 [-0.12, 0.28]$; gte 0.00	holds
P3 no drop: κ_R CI $\ni 0$	$0.019 [0.003, 0.034]$; gte 0.017	failed
P4 drop at $R = 0$	no drop; $R = 0$: -0.10	moot
P5 $\delta_V \approx \delta_F$	-0.04 vs -0.08	holds
P6 dedupe adds a drop	$+0.005 / -0.03$	≈ 0
#38 (powered)	$-0.26 [-0.38, -0.07]$: a rise	failed
51g (powered)	$0.05 [-0.13, 0.19]$	failed



Second figure. Pooled event profile: room alignment of the last three statements before and the first ten after a forced erasure, divided by the pre-erasure mean (95% CIs). No dip at the erasure; the later climb is post hoc and mixes segment lengths (later positions come only from long segments).

GOAL PERIODS IN DETAIL

G36 d descriptive
 G37 ~ mixed
 G38 x failed
 G39 ~ mixed
 G41 ~ mixed
 G42 ~ mixed
 G44 ~ mixed
 G51 x failed
 NE41 x failed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`witeup/paper.tex`).

CAVEATS

Scoped statements are sparse (median 2 per reset segment), so effect sizes are read on the estimand scale (Amendment 1); HH339's literal "30% for 3 statements" world has an estimand of only 0.12. Per-period power is ≥ 0.8 only in #38 and 51g; six per-period negatives are inconclusive. The room axis is a period-level ruler; a faster, other-direction room component would be missed. A drop that recovers before the first post-erasure statement is invisible. The post-erasure rise ($-0.12 [-0.21, -0.03]$ direct pooled) is post hoc.

ROUND 2

Test whether room alignment falls with context fill inside a segment (does context dilute room order?), and measure the drop on the call clock in the first 1–3 calls after an erasure. Confirm on #45–#47 and #50.

- **H109-R1** What holds the order?
- **H109-R2** The post-erasure rise.
- **H109-R3** A call-clock version.
- **H109-R4** Confirm on #45–#47, #50 after Vivian's sign-off and the ledger disclosure.